

Homework 1 - Stat 495 - Fall 2025

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Homework 1 - Due Wednesday, Sept. 24th by midnight to Gradescope

Practicing Academic Integrity

If you worked with others or used resources outside of provided course material, such as Generative AI (anything besides our textbook, course materials in the repo, labs, R help menu) to help complete this assignment, please acknowledge them below using a bulleted list. Remember that according to our class framework, if you engage with Generative AI tools, the tool and date of use must be listed here, and then a complete record of the chat history must be appended to the assignment.

I acknowledge the following individuals with whom I worked on this assignment:

Name(s) and how they assisted you

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I used the following sources to help complete this assignment:

Source(s) and where used in your submission

- I used my 370 chapter 10 part 3 notes to confirm my response to additional 3.

For this assignment, you may do some of the problems on paper if you like, then scan in your solutions and merge with the Integrity page as a cover sheet and any work you are leaving in the .Qmd. If you decide to type in your answers, you will need to use LaTeX to show your work for some problems. Add just means Additional (not from the textbook).

Because you may merge files into a pdf to submit in any order, be sure to assign pages in Gradescope so I can find work for each problem!

PROBLEMS TO TURN IN: Add 1, CASI 2.1 (modified), Add 2, Add 3, CASI 5.3 (modified), Add 4, Add 5

Additional 1

Suppose you have a random sample of n observations drawn from a Bernoulli distribution with parameter θ . Further suppose that θ is unknown, but has a prior density of a $\text{Beta}(\alpha, \beta)$ distribution, with both α and β greater than 0.

part a: Derive the posterior density for theta given the data. Be sure to fully specify/identify the posterior density in your solution. **Derive** means you must show your computations, even if this is result known to you.

SOLUTION:

part b: Find the Bayesian estimator for theta (i.e. the posterior mean).

SOLUTION:

part c: Is this a conjugate prior setting? Explain why it is or why it is not.

SOLUTION:

CASI 2.1 (modified, added a part c)

A coin with probability of heads θ is independently flipped n times, after which θ is estimated by

$$\hat{\theta} = \frac{s+1}{n+2};$$

with s equal to the number of heads observed. $\hat{\theta}$ will be referred to as the estimator below.

- (a) What are the bias and variance of the estimator?

SOLUTION:

- (b) How would you apply the plug-in principle to get a practical estimate of the standard error of the estimator?

SOLUTION:

- (c) Suppose that $n = 18$ and $s = 6$. Compute the usual $\hat{p}$ used to estimate θ , the new estimator $\hat{\theta}$, and the SE of each estimator. Write a sentence or two comparing the estimators addressing the following questions: How similar are the estimators? How similar are their SEs? Which do you prefer? Why?

SOLUTION:

(Bonus, not worth any points) $\hat{\theta}$ is the estimator that would result in a Bayesian context under a certain prior for p . Can you identify which one (all details)?

Additional 2

Suppose $X_1 \dots X_n$ are iid from an Exponential distribution with parameter β , with pdf:

$$f(x|\beta) = \frac{1}{\beta} \exp^{-x/\beta}, x > 0, \beta > 0,$$

and 0, otherwise.

- (a) Find the MLE for beta.

SOLUTION:

- (b) Verify the MLE is unbiased.

SOLUTION:

- (c) Find the Fisher information for a single observation.

SOLUTION:

- (d) Find the Cramer Rao lower bound on variance for unbiased estimators of beta.

SOLUTION:

- (e) MLEs have excellent properties, but there is one downside to / problem with them that was introduced in the early CASI chapters. What is the potential issue? Hint: it became evident as data sets got bigger.

SOLUTION:

Additional 3

In a few sentences and in your own words, explain what the Neyman-Pearson lemma tells us and why it is important/useful in the context of hypothesis testing.

SOLUTION:

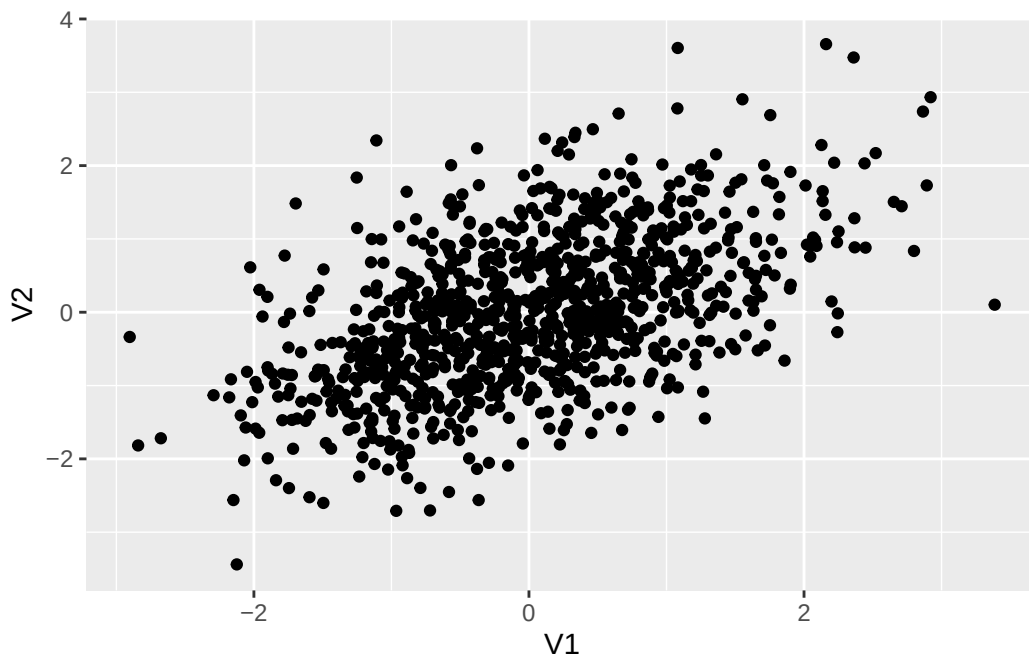
CASI 5.3 (modified)

Draw a sample of 1000 bivariate normal vectors $x = (x_1, x_2)'$, with each variable having a mean of 0, a standard deviation of 1, and with a correlation between them of 0.5. Be sure your process is reproducible. Review the chapter 4 and 5 practice problems for assistance with code.

(a) Plot your sample of data points.

SOLUTION:

```
set.seed(495)
cov <- 0.5 * 1 * 1
sigma <- matrix(c(1, cov, cov, 1), ncol = 2)
NormData <- as.data.frame(rmvnorm(1000, mean = c(0, 0), sigma = sigma))
gf_point(V2 ~ V1, data = NormData)
```



(b) Following equation 5.19, what should the theoretical distribution of $x_2|x_1$ be here?

SOLUTION:

$$x_2|x_1 \sim N\left(\mu_2 + \frac{\sigma_{12}}{\sigma_{11}}(x_1 - \mu_1), \sigma_{22} - \frac{\sigma_{12}^2}{\sigma_{11}}\right)$$

$$\Rightarrow x_2|x_1 \sim N\left(0 + 0.5(x_1 - 0), 1 - \frac{0.25}{1}\right) = N(0.5x_1, 0.75)$$

(c) Regress x_2 on x_1 and numerically check equation 5.19.

Hint: Reading the text will help you understand what two things to check in the regression output.

SOLUTION:

```
lm1 <- lm(V2 ~ V1, data = NormData)
summary(lm1)
```

Call:

```
lm(formula = V2 ~ V1, data = NormData)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.3585	-0.5575	-0.0282	0.5476	3.0153

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.00323	0.02747	-0.12	0.91
V1	0.54744	0.02800	19.55	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.869 on 998 degrees of freedom

Multiple R-squared: 0.277, Adjusted R-squared: 0.276

F-statistic: 382 on 1 and 998 DF, p-value: <2e-16

Linear regression coefficient:

The observed $\hat{\beta}_1 = 0.54744$ is very similar to the calculated slope ($\sigma_{12}/\sigma_{11} = 0.5$) with a very small difference of ~ 0.04 .

R^2 :

Similarly the observed $R^2 \approx 0.277$ is very similar to the calculated one ($\frac{\sigma_{12}^2}{\sigma_{11}\sigma_{22}} = 0.25$) with a very negligible difference of ~ 0.02 .

Additional 4

Many parametric inference procedures rely on certain conditions being met in order for the procedures to be valid. In cases where the conditions are not met, nonparametric procedures can be employed. You have seen the bootstrap and permutation/randomization tests as examples of alternatives (much more exists in the field of nonparametric statistics). In the practice problems, you saw an example of a permutation test on gene data.

How are these procedures adapted to other situations?

- (a) In a few sentences, describe how you would perform a permutation-based test to assess the overall significance of a multiple linear regression. If context would help, you can consider the following regression model:

```
data(penguins)
mymod <- lm(bill_length_mm ~ bill_depth_mm*species, data = penguins)
msummary(mymod)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	23.068	3.017	7.65	2.2e-13	***
bill_depth_mm	0.857	0.164	5.22	3.1e-07	***
speciesChinstrap	-9.640	5.715	-1.69	0.09259	.
speciesGentoo	-5.839	4.535	-1.29	0.19885	
bill_depth_mm:speciesChinstrap	1.065	0.310	3.44	0.00067	***
bill_depth_mm:speciesGentoo	1.164	0.279	4.17	3.8e-05	***

Residual standard error: 2.44 on 336 degrees of freedom

(2 observations deleted due to missingness)

Multiple R-squared: 0.802, Adjusted R-squared: 0.799

F-statistic: 273 on 5 and 336 DF, p-value: <2e-16

Note that there aren't necessarily problems with using the parametric methods in this setting, but context can be helpful for thinking through the process. You don't need to implement the procedure either, but feel free if it helps your description.

SOLUTION:

In this context I would:

1. Fit the regression of bill length on bill depth, species, and their interaction and record the observed F-statistic (273).
2. Under the null that bill length has no relationship with any of the predictors, randomly shuffle the bill length values among penguins while leaving the predictors fixed.
3. Refit the regression using the shuffled data and compute the new F-statistic.

4. Repeat the two steps above about 1000 times to make a null distribution of F-statistics. Plot the distribution.
5. Compare the observed F-statistic (273) to this null distribution and estimate the p-value. How unusual the obs. F-stat is under the null tells us if the predictors really explain variation in bill length or if the result could just be due to chance.

(b) Describe a way that a bootstrap might be useful in a multiple linear regression model.

Again, if context is useful, feel free to refer to the model fit above. There are many possible solutions here. As above, you don't need to implement the bootstrap, but feel free if it helps your description.

SOLUTION:

In this context I would:

1. Draw the same number of penguins as in the original dataset (342), but sample with replacement so some penguins may appear multiple times while others may not appear at all.
2. Fit the regression model on this bootstrap sample.
3. Record the estimates for all regression coefficients.
4. Repeat steps 1–3 above many times e.g. 1000.
5. Collect the estimates across the samples to form bootstrap sampling distributions for each regression coefficient.
6. Use these distributions to calculate confidence intervals for the coefficients and assess their significance.

Additional 5 - Project Check-In and Updated Timeline

The tailored project is due Friday, Oct. 3rd, leaving a little over a week between this assignment deadline and its due date. You will also have a Hmk 2 due next week on the new material in CASI on predictive modeling.

- (a) Briefly, summarize the status of your project. What have you completed?
What is left?

SOLUTION:

So far I have:

- Found and loaded and e-commerce dataset into R.
- Refreshed my knowledge of how Power BI works.
- Examined the data to understand its contents and all the variables.
- Completed the first round of cleaning and created new variables to make analysis easier.
- Made some basic plots to perform EDA
- Decided on my potential regression question.

What's left is building and checking the regression model in R, interpreting the results, preparing aggregated metrics for the dashboard, creating the interactive BI dashboard, and putting together the final report with all the analysis and insights.

- (b) Provide a suggested timeline/set of deadlines (whatever works best for you)
for what you have left so you can finish the project by the deadline.

SOLUTION:

- 09/24-09/25: Build regression models and finalize statistical analysis. - 09/26: Finalize Power BI dashboard. - 09/27-09/29: Do final write-up, integrate dashboard screenshots, finalize repo, and deliver the two final deliverables. - 10/01: Submit report, Power BI file, and all project files.

- (c) Any questions about the tailored project for me at this juncture?

SOLUTION:

No question.